

ACS Filterlab user guide

This section shows how to test the connection between the ACS and FilterLab, with instructions on how to set up a measurement in FilterLab.

The following instructions require the user to operate from a Windows system.

Launch the acs server on the PC or target device.

Then, start FilterLab and import the following configuration:

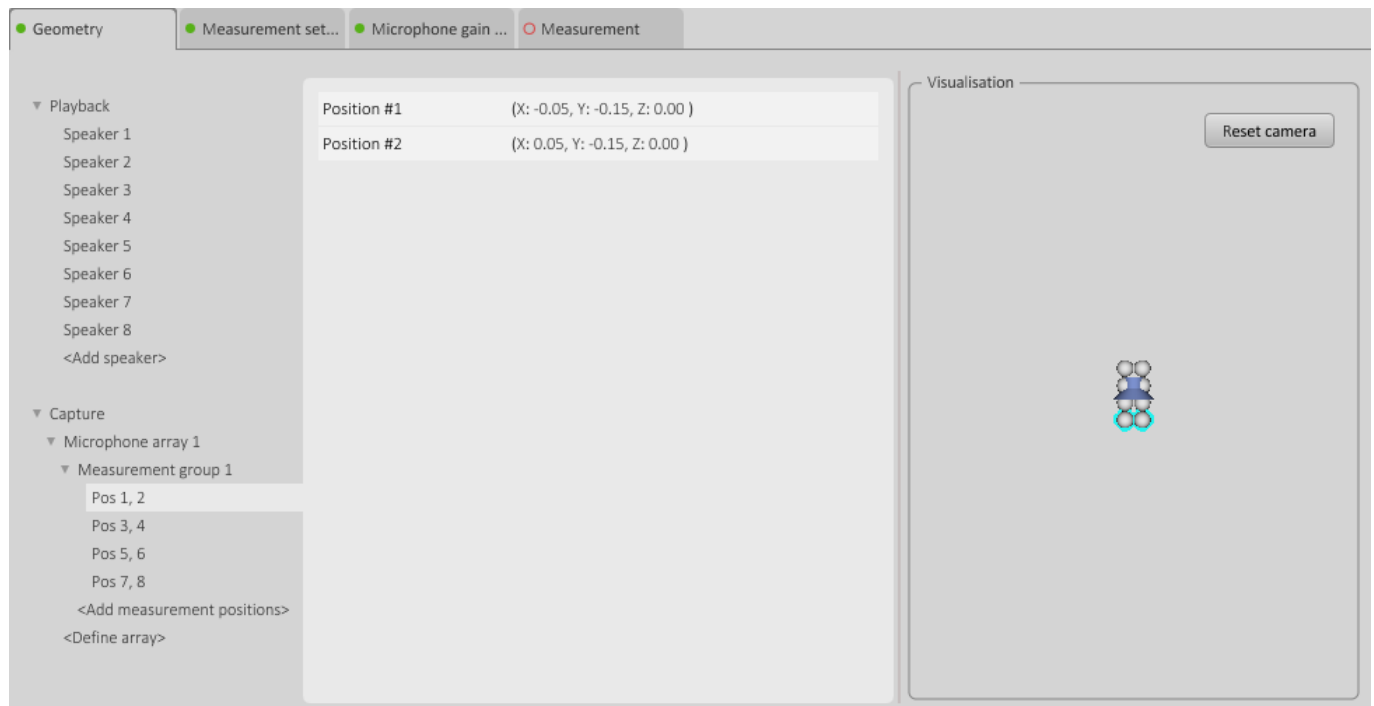
- `filterlab/measurement_audio_settings.configuration`

Then, double-click on the configuration that now appears on the under the **Configurations** table.

Then, from the menu **Import data**, select **Geometry** and import:

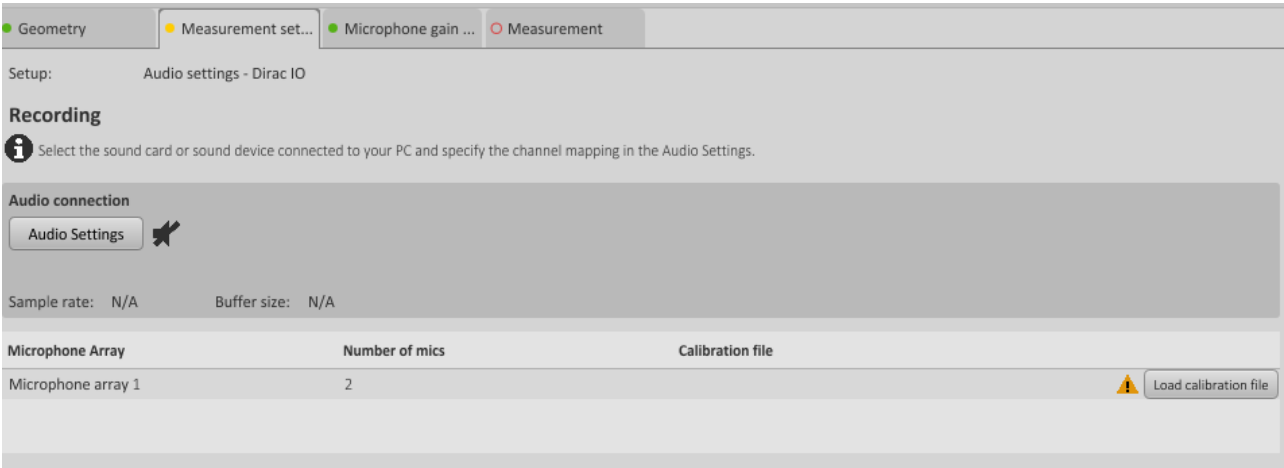
- `resources/filterlab/measurement_geometry.geometry`

As a result, the **Geometry** tab will be filled with a meaningful setup (see figure below).



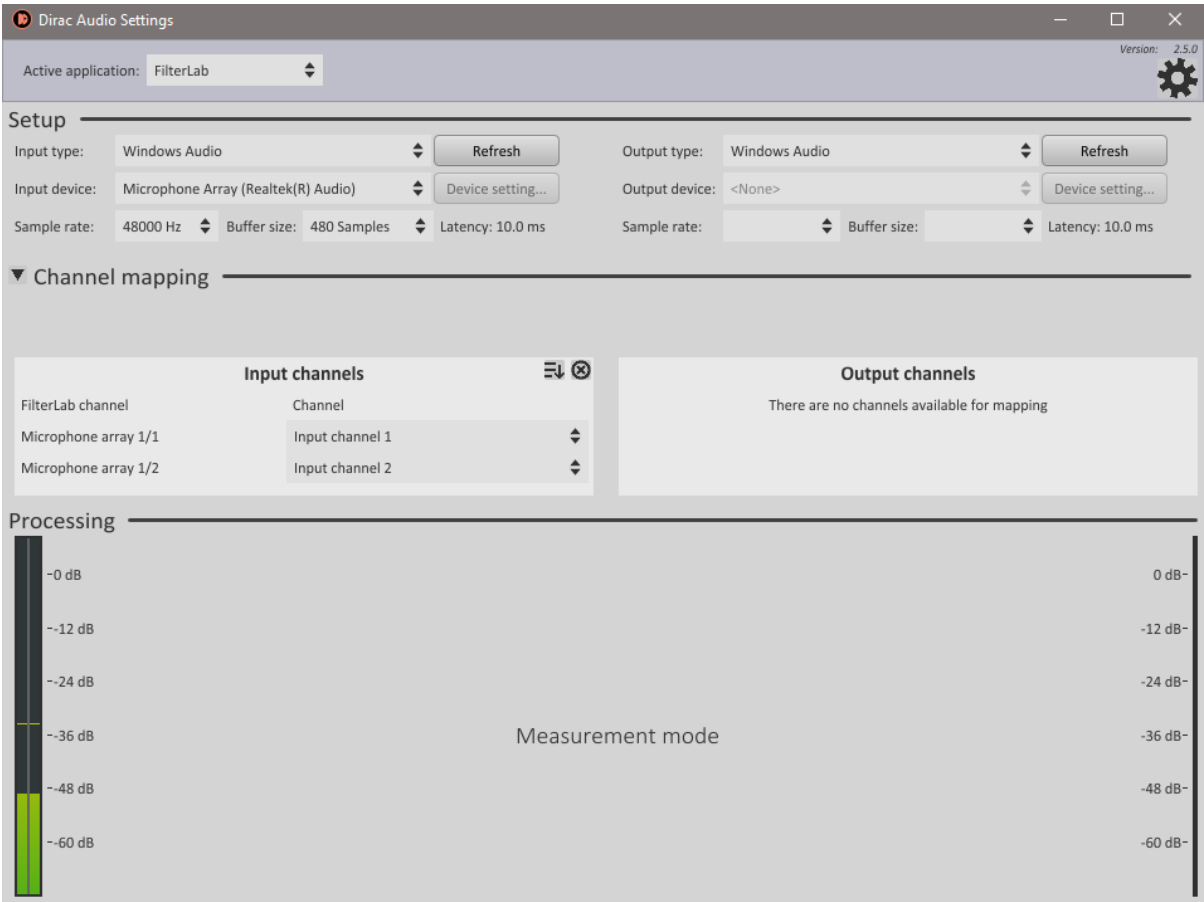
- **The **Measurement setup** tab**

In the **Recording** panel on the left, press the **Audio Settings** button:



This will open **Dirac Audio Settings** in a new window.

◦ **The Dirac Audio Settings window**



Under the **Setup** panel, choose **Windows Audio** as input type, then select whatever microphone you have on your system as **Input Device**.

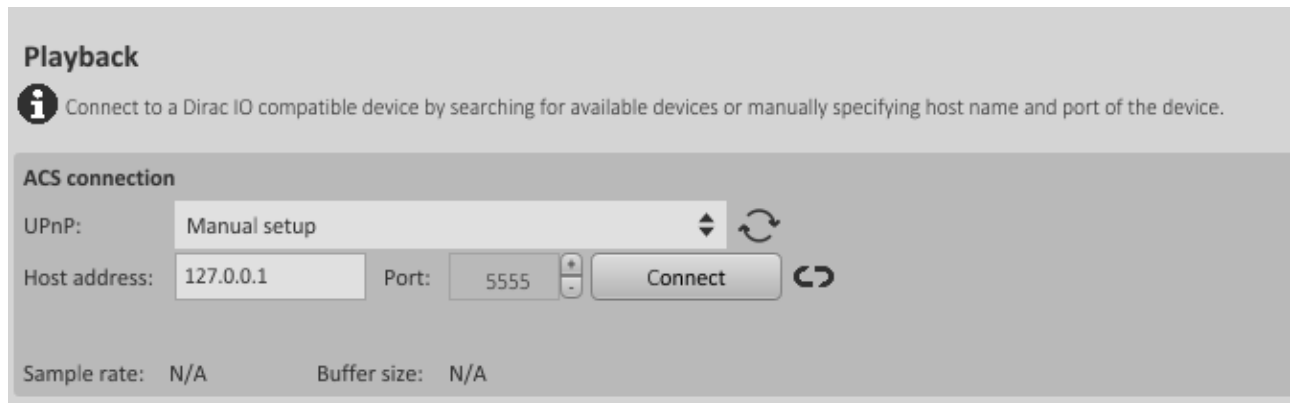
Under the **Channel mapping** panel, and specifically under the **Input channels** sub-panel, click the following button:



This will have the effect of directing the audio stream from your input device into FilterLab.

For the sake of only testing the communication between FilterLab and the ACS, you can disregard the output setup and minimize the **Dirac Audio Settings** window.

Going back to the main FilterLab window, in the **Playback** panel on the right, under the **ACS connection** sub-panel, make sure that the **Port** field matches the port selected when launching the ACS server (see above), then press the **Connect** button:



This will establish a connection between FilterLab (which will act as a client) and the ACS server:

- The connection icon to the right of the button will change accordingly (turning green).
- Some logging activity will be shown in the terminal where the ACS server is running (provided the **-v** option was used when launching the server).

At this point, you can play around with test signals:

- Select **Signal type** and **Gain** from the **Test signal settings** sub-panel.
- Play the selected stimulus on a given channel by pressing one of the **play** (black triangle) buttons.
 - Stop a playing stimulus by pressing the corresponding **stop** (black square) button.
- Select the marker channel in the **Speaker for detection sweep** menu at the bottom.

Test signal settings

Signal type: Pink noise

Gain: -10 dB

Speaker cutoffs will NOT be applied to the test sound

Measurement type

☒ Room measurement

☐ Speaker measurement

Speaker	Type	Lower cutoff	Test
Speaker 1	Mid/High	N/A	▶
Speaker 2	Mid/High	N/A	▶
Speaker 3	Mid/High	N/A	▶
Speaker 4	Mid/High	N/A	▶
Speaker 5	Mid/High	N/A	▶
Speaker 6	Mid/High	N/A	▶
Speaker 7	Mid/High	N/A	▶
Speaker 8	Mid/High	N/A	▶

Speaker for detection sweep: Speaker 4

While performing the above actions, you will see the corresponding activity (logs) in the terminal where the server is running.

• The Measurement tab

In the left panel, press one of the grey round buttons next to the measurement groups.

● Geometry

● Measurement set...

● Microphone gain ...

○ Measurement

▼ Microphone array 1

▼ Measurement group 1

Pos 1, 2

0 dB

▶

●

Pos 3, 4

0 dB

▶

●

Pos 5, 6

0 dB

▶

●

Pos 7, 8

0 dB

▶

●

This will trigger a measurement:

- In the bottom-right panel, you will see the incoming audio waveform as it is being captured by your microphone.
- In the terminal where the server is running, you will see a progress report as the stimuli are being played over all channels.